Design Document: Detecting Out-of-Context Image-Caption Misinformation

DATA5500 Senior Design | Fall 2026 | Instructor: Weijie Pang | Student: Mohamed Mohamed | Updated October 13, 2026 (originally submitted September 27, 2026)

Revision note (October 13, 2026): updated after completing data preparation, the embedding-generation experiment, and the similarity-threshold baseline. Changes are in Sections 2 (datasets, tools, infrastructure), 3 (two new decisions), 4 (two pending decisions resolved), 7 (Weeks 3 to 6 marked complete), and 8 (risks updated with measured results).

1. Overview of the Project, Including Its Potential Impact and Market

This project builds a system that detects out-of-context (OOC) misinformation: cases where a real, unaltered photograph is paired with a caption that misrepresents what it actually shows. Using CLIP, a pretrained model that embeds images and text into a shared vector space, the system measures whether a given image and caption are semantically consistent, then classifies the pair as a likely genuine match or a likely out-of-context mismatch. The project spans the full pipeline: acquiring and structuring benchmark data, generating embeddings, training and comparing classifiers of increasing sophistication, evaluating performance by mismatch type, and deploying the result as a live, interactive application.

The potential impact reaches several audiences. Fact-checking organizations and journalists currently rely on manual investigation or general-purpose reverse-image search to catch this specific misinformation pattern; a lightweight, automated first-pass signal could help them triage claims faster. Social media platforms and content moderation teams face the same problem at a much larger scale, where even a partial automated signal has real value. There is also a media literacy and education angle: a public-facing demo makes an otherwise abstract concept, that a caption can lie even when a photo is completely real, immediately tangible to a general audience. Within the scope of this course, the project also serves as a portfolio piece demonstrating an end-to-end multimodal machine learning system.

2. Existing Datasets, Resources, Tools, and Infrastructure

Datasets

- **NewsCLIPpings** (github.com/g-luo/news_clippings, also at huggingface.co/g-luo/news-clippings, a Hugging Face model repository): the primary benchmark, providing labeled genuine and falsified image-caption pairs across defined mismatch-generation-method subsets (person-match, scene-match, and semantic/CLIP-based match). It also ships precomputed CLIP ViT-B/32 image and text embeddings for every sample. *Status: downloaded and validated (2.6 GB); a balanced 24,000-pair subset is built.*
- **VisualNews** (github.com/FuxiaoLiu/VisualNews-Repository): the underlying news image-caption corpus that NewsCLIPpings draws from. It is an open download (no access request), distributed as one 91 GB archive whose server supports HTTP range requests, so a partial download yields usable images. NewsCLIPpings records map to VisualNews images through the metadata image_path field. *Status: 1.5 GB partial download used for the embedding-generation experiment.*
- **COSMOS** (github.com/shivangi-aneja/COSMOS): a related out-of-context misinformation benchmark, planned as an optional secondary check on whether findings generalize beyond NewsCLIPpings' specific mismatch-generation methodology.

Tools

- Python for the full pipeline: data handling, embedding generation, modeling, and the application layer.
- OpenAI's CLIP model (openai/clip-vit-base-patch32 via Hugging Face Transformers), used off-the-shelf through transfer learning rather than trained from scratch.
- scikit-learn for the logistic regression baseline and standard evaluation metrics; PyTorch for the lightweight neural network classifier.
- Pandas with Parquet files for the curated subset.
- Streamlit for the deployed, interactive demo application.
- Matplotlib and/or Plotly for the evaluation dashboard's charts and the embedding-space visualization.
- Jupyter notebooks for exploratory development and experimentation.

Infrastructure

- Local machine (Apple Silicon laptop with the PyTorch MPS backend) for development, embedding generation, modeling, and running the Streamlit app during testing. Measured throughput is 224 images per second end to end, so Google Colab is no longer needed for embedding generation; it remains a fallback.
- GitHub for version control and as the reproducible record of the project's code and documentation.
- A public hosting option for the live demo (Streamlit Community Cloud or Hugging Face Spaces) is under consideration; see Pending Design Decisions.

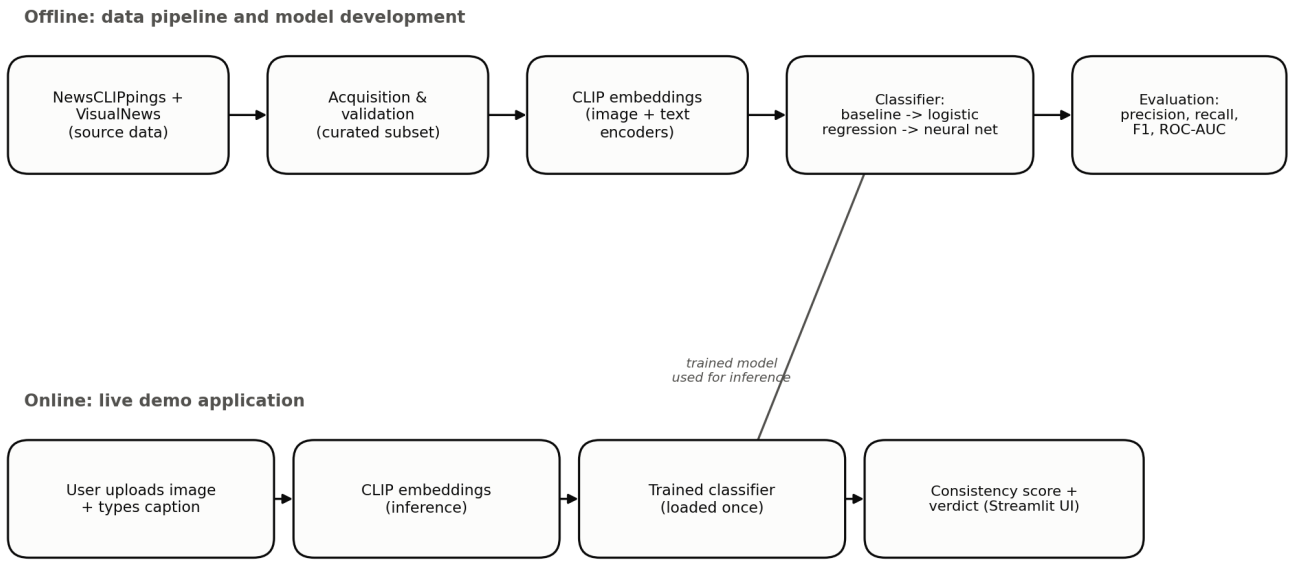


Figure 1. System architecture and data flow. The top row is the offline pipeline that produces a trained classifier; the bottom row is the online path a user's submission takes through the deployed application, using that same trained classifier for inference.

3. Key Design Decisions Already Made

Detection approach: embedding similarity over pixel forensics

Because out-of-context misinformation uses a genuine, unaltered image, the project is built around comparing image and caption meaning (via CLIP embeddings) rather than looking for pixel-level manipulation, which would find nothing wrong with these images by design.

Primary dataset: NewsCLIPpings, not custom scraping

NewsCLIPpings was chosen as the core dataset instead of scraping live news or social media content. This avoids the acquisition reliability and terms-of-service risk that comes with scraping, and provides an established, labeled benchmark with a defined evaluation methodology.

Curated subset: 24,000 balanced pairs (new, October 13)

The working dataset is 24,000 pairs sampled from NewsCLIPpings' merged_balanced split (19,200 train / 2,400 validation / 2,400 test), with an equal count in every mismatch-method and label cell (600 test pairs per method). This keeps per-method test results statistically usable (bootstrap 95% confidence intervals of roughly plus or minus 0.04 AUC per method) while staying small enough to iterate quickly.

Embeddings: provided CLIP embeddings for modeling, own pipeline for inference (new, October 13)

Models are trained on the CLIP ViT-B/32 embeddings that NewsCLIPpings provides, so modeling does not depend on downloading all 91 GB of VisualNews images. The embedding pipeline the app uses was validated against them on 2,641 real images: mean cosine agreement is 0.9936 for images and 0.999999 for captions.

Modeling approach: staged, transfer-learning based

The classifier is being developed in stages: a similarity-threshold baseline (complete, test ROC-AUC 0.716), then a logistic regression model, then a lightweight neural network, all built on top of pretrained CLIP embeddings rather than training a

vision-language model from scratch. This keeps the technical scope realistic for one student in one semester.

Evaluation approach: per-category breakdown as a first-class result

Rather than reporting a single aggregate accuracy figure, the evaluation plan treats performance by mismatch-generation method (person-match, scene-match, and semantic/CLIP-based match) as a primary result. The baseline confirmed this matters: test ROC-AUC ranges from 0.910 on the CLIP text-text split to 0.458 on the CLIP text-image split.

Deployment approach: Streamlit with preset examples plus user upload

Streamlit was chosen for the live demo because it allows a working interactive interface to be built quickly by a solo developer. The interface will support both user-uploaded images and a small set of preset example pairs, so the demo works even for a visitor without an image on hand.

Scope boundary: a decision-support signal, not a verdict

The tool will be framed and documented as a decision-support signal (a consistency score with an explanation), not as a definitive true/false judgment, to avoid overstating the reliability of an embedding-similarity approach.

4. Pending Design Decisions

- Whether to run the COSMOS dataset as a secondary generalization check, or keep the evaluation scope to NewsCLIPpings only if time becomes limited.
- Whether to add a retrieval-based explainability feature (showing a visually similar genuine pair alongside a flagged result), or keep the interface to a score and verdict only.
- Where to host the live demo for the final presentation and poster session: running locally on a laptop versus a public deployment on Streamlit Community Cloud or Hugging Face Spaces.

Resolved since September 27

- **Curated subset size:** 24,000 pairs (see Section 3).
- **Broken or dead image URLs:** not applicable. NewsCLIPpings annotations reference local VisualNews ids that map to image files inside the VisualNews archive, not web URLs. This only becomes relevant again if COSMOS or URL input in the app is added.

5. Project Goals for Academic Success

This project is designed to demonstrate the full data science lifecycle covered by the program: real data acquisition and validation, data management and preparation, multimodal machine learning analysis, staged model development with rigorous statistical evaluation, and deployment with visualization. It applies coursework in machine learning, statistics, and software development to a well-defined, benchmarked problem, and produces a defensible, evaluated result rather than a purely illustrative demo.

6. Additional Goals for Real-World Application

Beyond the course requirements, the finished system is intended to work as a genuine, usable proof-of-concept: something a fact-checker, journalist, or curious member of the public could actually try, not only read about. The completed code, documentation, and live demo will also serve as a portfolio artifact demonstrating applied multimodal machine learning, and the documented pipeline is written so it could plausibly be extended after the semester, for example into a browser extension or a batch-checking tool for a larger set of images.

7. Updated Project Plan and Timeline

This follows the same 14-week structure submitted with the Project Plan, with Week 1 beginning September 13, 2026. Weeks 1 to 6 are marked complete below. Dates remain planning estimates and will be reconciled against the official course calendar as deadlines are confirmed.

Week	Milestone / Deliverable	Activities
1	Project kickoff & refinement	Review the approved proposal; finalize the project objectives and the CLIP-embedding-based detection methodology. (Completed.)

Week	Milestone / Deliverable	Activities
2	Dataset exploration & acquisition	Identify and begin acquiring data from NewsCLIPpings and VisualNews; review COSMOS as a candidate secondary dataset. Project Plan submitted this week. (Completed.)
3	Data preprocessing	Download and validate NewsCLIPpings; build the balanced 24,000-pair subset with train/val/test splits. Design Document submitted. (Completed.)
4	Exploratory embedding analysis	Generate CLIP embeddings for 2,641 VisualNews images and captions and validate them against the provided embeddings; analyze similarity distributions for genuine and falsified pairs. (Completed.)
5	Baseline modeling	Similarity-threshold baseline: test ROC-AUC 0.716 overall, 0.458 to 0.910 by method. (Completed.)
6	Visualization of categories and trends	Similarity histograms by method, ROC curve, per-method AUC/F1 chart, PCA and t-SNE embedding-space plots. Progress Report 2 submitted. (Completed.)
7	Midterm progress report	Submit a summary of findings so far; receive instructor feedback on baseline results and modeling direction.
8	Classifier development - Part I	Train and tune a logistic regression classifier on the CLIP embeddings; compare performance to the baseline.
9	Classifier development - Part II	Train and tune a lightweight neural network classifier; perform a per-category performance gap analysis against the logistic regression model.
10	Optional: secondary generalization check	If time allows, evaluate the trained classifier on the COSMOS dataset to test generalization beyond NewsCLIPpings' mismatch-generation methodology.
11	Application interface development	Build out the Streamlit demo interface (image upload, preset examples, results panel) and integrate the best-performing trained classifier for live inference.
12	Report writing & visual storytelling	Draft the final report and documentation; refine the evaluation dashboard and embedding visualizations.
13	Slide deck preparation & report revision	Create the final presentation slides and poster; finalize the written documentation and code repository.
14	Final presentation & submission	Deliver the final presentation and submit the Final Project and Documentation.

8. Risks, Unknowns, and Concerns

- **VisualNews download size (replaces the earlier link-rot risk).** Link rot does not apply, since images come from the VisualNews archive rather than live URLs. The new constraint is that the 91 GB archive downloads at about 1.7 MB/s (roughly 15 hours). Mitigation: modeling uses the provided embeddings, and range requests let me pull only the images I need for the experiment and the demo's preset examples.
- **Compute and time constraints (largely resolved).** Measured local throughput is 224 images per second end to end, so embedding the full subset takes minutes. Colab remains a fallback.
- **Preprocessing mismatch between training and the live app (new).** My own image embeddings agree with the provided ones at mean cosine 0.9936 but drop as low as 0.901 for a few images, likely due to resize and crop differences (not yet verified). Mitigation: confirm the cause and match the app's preprocessing to the training embeddings before Week 11.
- **Generalization limits.** NewsCLIPpings' falsified pairs are synthetically generated by automatic retrieval, which may not fully represent how out-of-context misinformation looks in the wild. This will be stated explicitly as a limitation rather than implied to be solved.
- **Category difficulty imbalance (confirmed).** The baseline scores 0.910 AUC on the CLIP text-text split but 0.458, below chance, on the CLIP text-image split, whose mismatched pairs were selected for high CLIP similarity. A model can look acceptable in aggregate (0.716) while failing on this split, which is why per-method evaluation stays a first-class result.
- **Schedule risk from optional features.** The pending decisions around explainability and public hosting are nice-to-have rather than core requirements. If the core pipeline falls behind schedule, these will be the first things deprioritized, given that no late work is accepted in this course.